

$$A = (t + 2m)(b + 2m)$$

$$10.125 = (0.60 + 2m)(0.30 + 2m)$$

$$10.125 = 0.18 + 1.20m + 0.60m + 4m^2$$

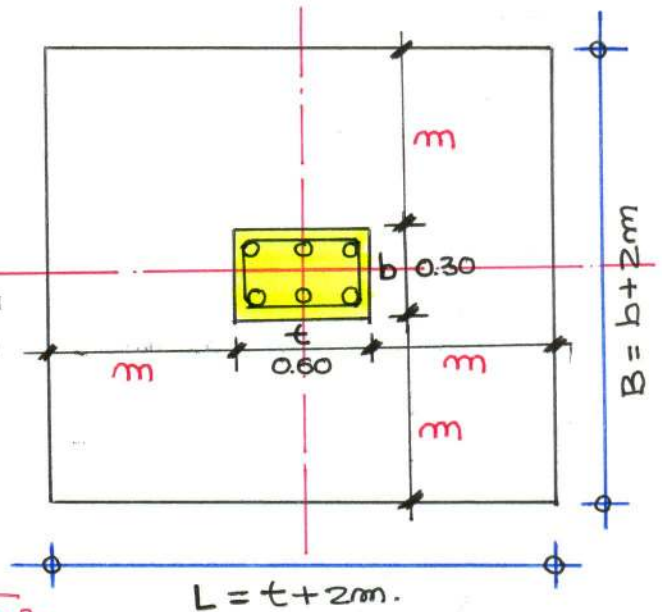
$$4m^2 + 1.8m - 9.945 = 0$$

$$m = 1.368 \text{ m.}$$

$$L = t + 2m = 0.60 + 2(1.368) \Rightarrow L = 3.336 \text{ m}$$

$$B = b + 2m = 0.30 + 2(1.368) \Rightarrow B = 3.036 \text{ m}$$

$$\therefore L = 3.30, B = 3.00, A = 9.90 \text{ m}^2$$



VERIFICACIÓN DE PRESIÓN $\sigma_{max} < \sigma_7$

$$P_s = P_{ext} + P_{ex} = 92.50 + 73.00 \Rightarrow P_s = 165.50 \text{ ton}$$

$$M_s = M_{ext} + M_{ex} = 36.50 + 22.50 \Rightarrow M_s = 59.00 \text{ ton-m}$$

$$\sigma_{max} = \frac{P_s}{A} + \frac{M_s c}{I}$$

$$A = 9.90 \text{ m}^2$$

$$c = L/2 \Rightarrow c = 1.65 \text{ m.}$$

$$I = \frac{BL^3}{12} \Rightarrow I = 8.984 \text{ m}^4$$

$$\sigma_{max} = \frac{165.50}{9.90} + \frac{59.00(1.65)}{8.984}$$

$$\sigma_{max} = 27.553 \text{ ton/m}^2 > \sigma_7 = 21.25 \text{ ton/m}^2 \quad (\text{No cumple la condición})$$

REDIMENSIONAMOS BY L

$$B = 3.40 \text{ m}, L = 3.70 \text{ m}, A = 12.58 \text{ m}^2, c = 1.85 \text{ m}, I = 14.352 \text{ m}^4$$

$$\sigma_{max} = \frac{P_s}{A} + \frac{M_s c}{I}$$

$$\sigma_{max} = \frac{165.50}{12.58} + \frac{59.00(1.85)}{14.352} \Rightarrow \sigma_{max} = 20.761 \text{ ton/m}^2$$

$$\sigma_{max} = 20.761 \text{ ton/m}^2 < \sigma_7 = 21.25 \text{ ton/m}^2$$

(CONFORME)

CARGAS DE DISEÑO (P_u, M_u)REACCIÓN AMPLIFICADA DEL SUELO:

$$P_{CM} = 92.50 \text{ ton} \quad M_{CM} = 36.50 \text{ ton-m.}$$

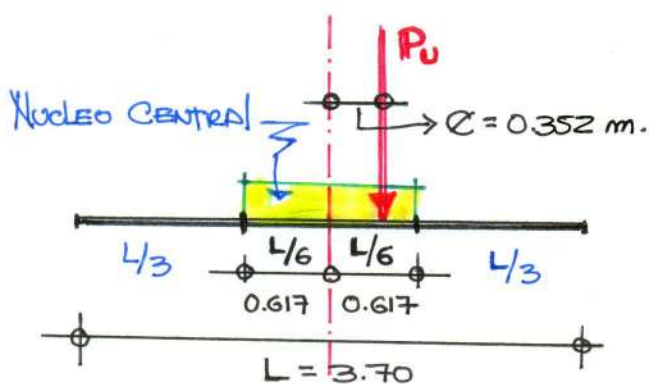
$$P_{CV} = 73.00 \text{ ton} \quad M_{CV} = 22.50 \text{ ton-m.}$$

$$P_u = 1.70 P_{CV} + 1.40 P_{CM} = 1.70(73.00) + 1.40(92.50) \Rightarrow 253.60 \text{ ton}$$

$$M_u = 1.70 M_{CV} + 1.40 M_{CM} = 1.70(22.50) + 1.40(36.50) \Rightarrow 89.35 \text{ ton-m.}$$

$$h = \# * d \Rightarrow e_u = P_u * e \Rightarrow e = \frac{M_u}{P_u} = \frac{89.35}{253.60} \Rightarrow e = 0.352 \text{ m.}$$

$$\frac{L}{6} = \frac{3.70}{6} \Rightarrow \frac{L}{6} = 0.617 \text{ m} \quad \because e = 0.352 < \frac{L}{6} = 0.617$$



Presión de contacto forma TRAPEZOIDAL

Presiones para el diseño (Método de Resistencia Última)

$$\sigma_{1,2} = \frac{P_u}{A} \pm \frac{M_u * c}{I}$$

$$\sigma_{1,2} = \frac{253.60}{12.58} \pm \frac{89.35(1.85)}{14.352}$$

$$\sigma_{1,2} = \begin{cases} \sigma_1 = 31.676 \text{ ton/m}^2 \\ \sigma_2 = 8.642 \text{ ton/m}^2 \end{cases}$$

$$P_u = 253.60 \text{ ton}$$

$$M_u = 89.35 \text{ ton-m.}$$

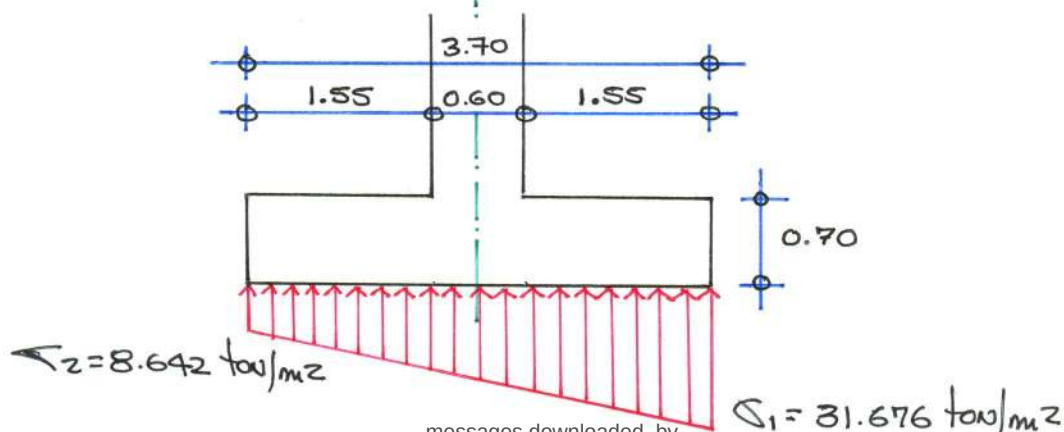
$$L = 3.70 \text{ m}$$

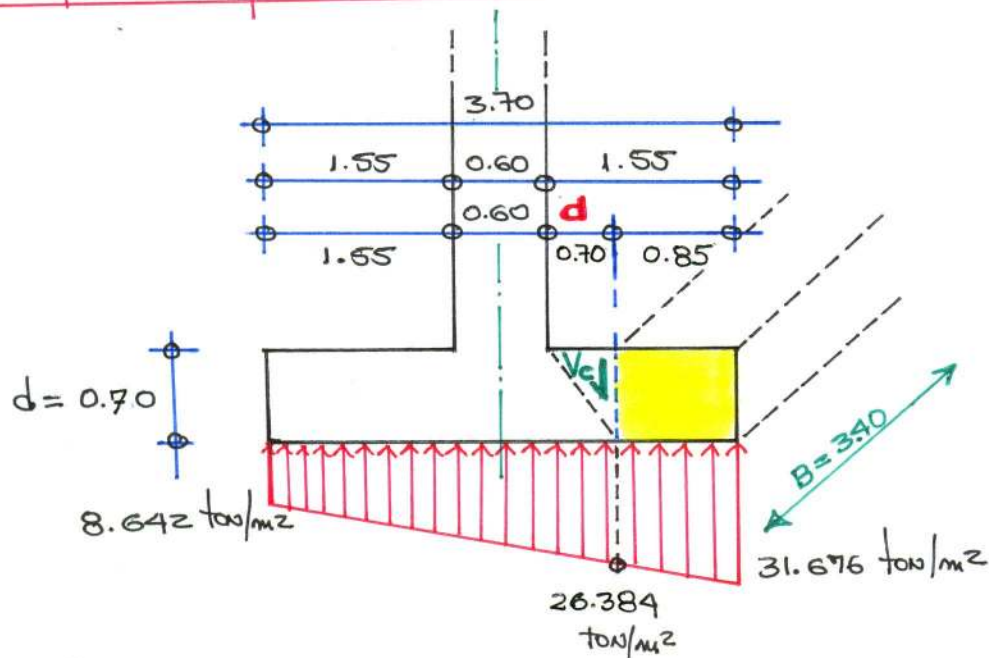
$$B = 3.40 \text{ m.}$$

$$A = 12.58 \text{ m}^2$$

$$c = 1.85 \text{ m}$$

$$I = 14.352 \text{ m}^4$$



VERIFICACION POR CORTANTE

$$V_u \leq \phi V_c$$

FUERZA CORTANTE ULTIMA $\longleftrightarrow$ RESISTENCIA DEL CONCRETO @ CORTE

$$V_u = \left[\frac{(31.676 + 26.384) \cdot 0.85}{2} \right] \cdot 3.40$$

$$V_u = 83.897 \text{ ton} = 83896.700 \text{ kg}$$

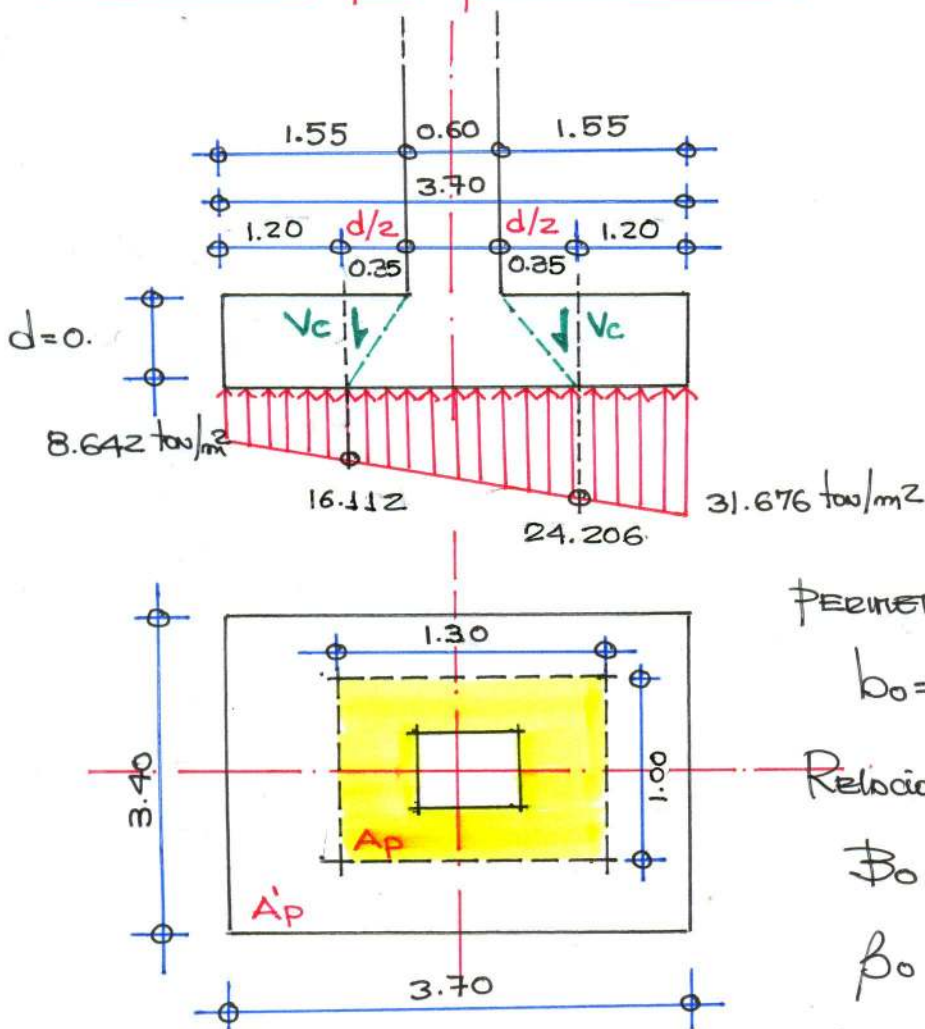
$$\phi V_c = \phi 0.53 \sqrt{f_c} B \cdot d$$

$$\phi V_c = 0.85 \cdot 0.53 \sqrt{210} \cdot 340 \cdot 70$$

$$\phi V_c = 155.375 \text{ ton} = 155375.092 \text{ kg}$$

$$\therefore V_u = 83.897 \text{ ton} < \phi V_c = 155.375 \text{ ton}$$

CONFORME

VERIFICACIÓN POR PUNZONAMIENTO

PERIMETRO DE PUNZONAMIENTO

$$b_o = 2(1.30) + 2(1.00) \Rightarrow b_o = 4.60 \text{ m}$$

Relación lados de columna

$$\beta_o = \frac{\text{lado mayor col.}}{\text{lado menor col.}}$$

$$\beta_o = \frac{0.60}{0.20} \Rightarrow \beta_o = 3$$

$$A'_p = A_z - A_p$$

$$A'_p = (370 \times 340) - (130 \times 100)$$

$$A'_p = 112800.000 \text{ cm}^2$$

Fuerza Cortante
Ultima

$$V_u \leq \phi V_c$$

Resistencia del
Concreto @ corte
Punzonamiento

$$V_u = q_u A'_p$$

$$\phi V_c = \phi \left(0.53 + \frac{1.10}{\beta_o} \right) \sqrt{f_c} b_o d$$

$$V_u = \left[\frac{(31.676 + 8.642) \cdot 3.70 \cdot (3.40)}{2} \right] - \left[\frac{(24.206 + 16.112) \cdot 1.30 \cdot (1.00)}{2} \right]$$

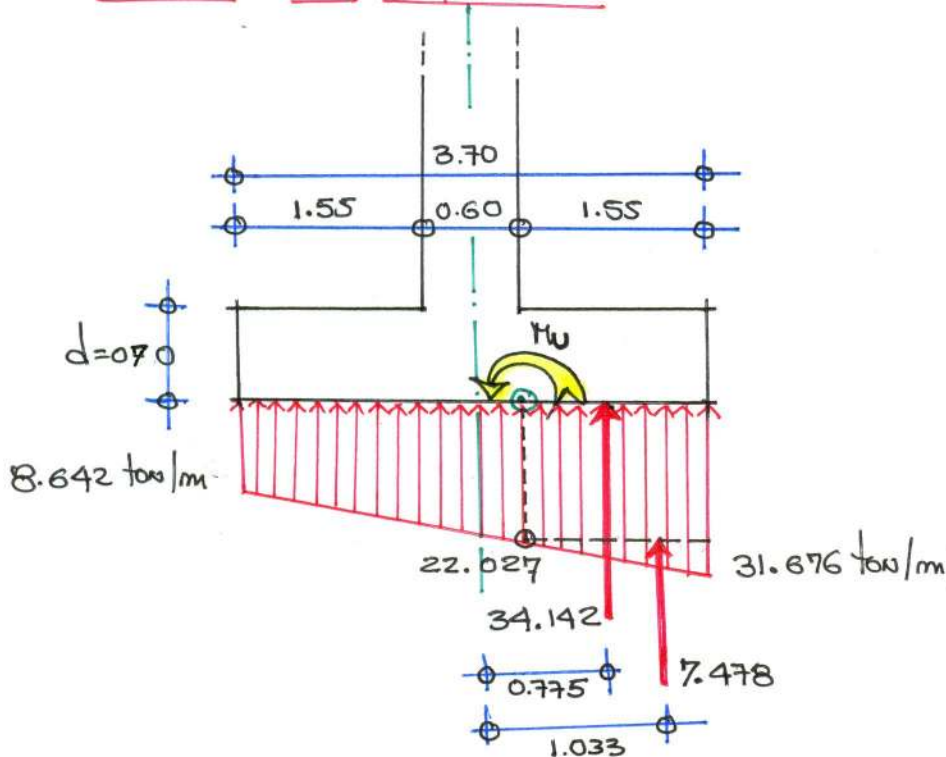
$$V_u = 227.394 \text{ ton} = 227393.520 \text{ kg}$$

$$\phi V_c = 0.85 \left(0.53 + \frac{1.10}{3} \right) \sqrt{210} \cdot 4.60 \cdot 0.70$$

$$\phi V_c = 428.359 \text{ ton} = 428359.300 \text{ kg}$$

$$V_u = 227.394 \text{ ton} < \phi V_c = 428.359 \text{ ton}$$

CONFIRME

DISEÑO DE REFUERZO:

$$M_u = 34.142(0.775) + 7.478(1.033) \Rightarrow M_u = 34.185 \text{ ton-m}$$

DISEÑO DE REFUERZO LONGITUDINAL

DATOS DE DISEÑO:

$$M_u = 34.185 \text{ ton-m}, b = 100 \text{ cm}, d = 70 \text{ cm}, f_c = 210 \text{ kg/cm}^2, f_y = 4200 \text{ kg/cm}^2$$

$$a = \frac{A_s f_y}{0.85 f_c b}$$

$$M_u = \phi A_s f_y (d - \frac{a}{2})$$

$$\rho_{min} = \frac{0.70 \sqrt{f_c}}{f_y} \Rightarrow \rho_{min} = 0.00242 \Rightarrow A_{smin} = \rho_{min} b d \Rightarrow A_{smin} = 16.906 \text{ cm}^2$$

$$\rho_b = 0.85 \beta_1 \frac{f_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) \Rightarrow \rho_b = 0.02125 \Rightarrow A_{sb} = \rho_b b d \Rightarrow A_{sb} = 148.75 \text{ cm}^2$$

$$\rho_{max} = 0.75 \rho_b \Rightarrow \rho_{max} = 0.01594 \Rightarrow A_{smax} = \rho_{max} b d \Rightarrow A_{smax} = 111.563 \text{ cm}^2$$

$$\rho_b = \omega \frac{f_c}{f_y} \Rightarrow \rho_b = 0.00189 \Rightarrow A_{sd} = \rho_b b d \Rightarrow A_{sd} = 13.214 \text{ cm}^2$$

$$A_s \phi 3/4 = 2.84 \text{ cm}^2 \quad d_b \phi 3/4 = 1.905 \text{ cm} = 0.01905 \text{ m}$$

$$A_{sd} < A_{smin} \Rightarrow A_{sd} = A_{smin} = 16.906 \text{ cm}^2$$

Numero de Varillas

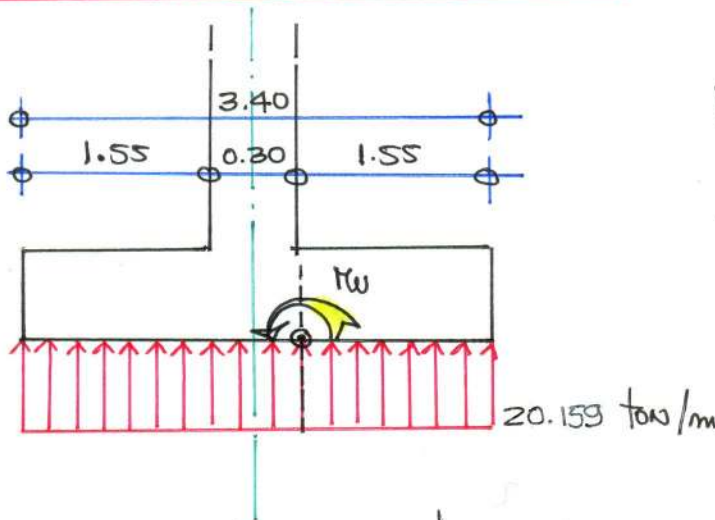
$$A_{st,d} = A_{st,d} (3.40) \Rightarrow A_{st,d} = 57.480 \text{ cm}^2$$

$$N = \frac{A_{st,d}}{A_{s\phi}} \Rightarrow N = \frac{57.480}{2.84} \Rightarrow N = 20.24 \Rightarrow N = 20 \text{ UNIDADES.}$$

Distribucion de Acero:

$$S = \frac{B - 2r - \phi - 0.10}{N-1} \Rightarrow S = \frac{3.40 - 2(0.075) - 0.01905 - 0.10}{1} \Rightarrow S = 0.165 \text{ m}$$

∴ Acero longitudinal: 20 $\phi 3/4 @ 0.165 \text{ m.}$

DISEÑO DE ACERO TRANSVERSAL

$$M_u = \frac{q L^2}{2} = \frac{20.159 (1.55)^2}{2}$$

$$M_u = 24.216 \text{ ton-m}$$

DATOS DE DISEÑO: $M_u = 24.216 \text{ ton-m}$, $b = 100 \text{ cm}$, $d = 70 \text{ cm}$, $f_c = 210 \text{ kg/cm}^2$, $f_y = 4200 \text{ kg/cm}^2$

$$P_{min} = \frac{0.70 \sqrt{f_c}}{f_y} \Rightarrow P_{min} = 0.00242 \Rightarrow A_{st,min} = P_{min} b d \Rightarrow A_{st,min} = 16.90 \text{ cm}^2$$

$$P_b = 0.85 \beta_1 \frac{f_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) \Rightarrow P_b = 0.02125 \Rightarrow A_{s,b} = P_b b d \Rightarrow A_{s,b} = 148.75 \text{ cm}^2$$

$$P_{max} = 0.75 P_b \Rightarrow P_{max} = 0.01594 \Rightarrow A_{s,max} = P_{max} b d \Rightarrow A_{s,max} = 111.56 \text{ cm}^2$$

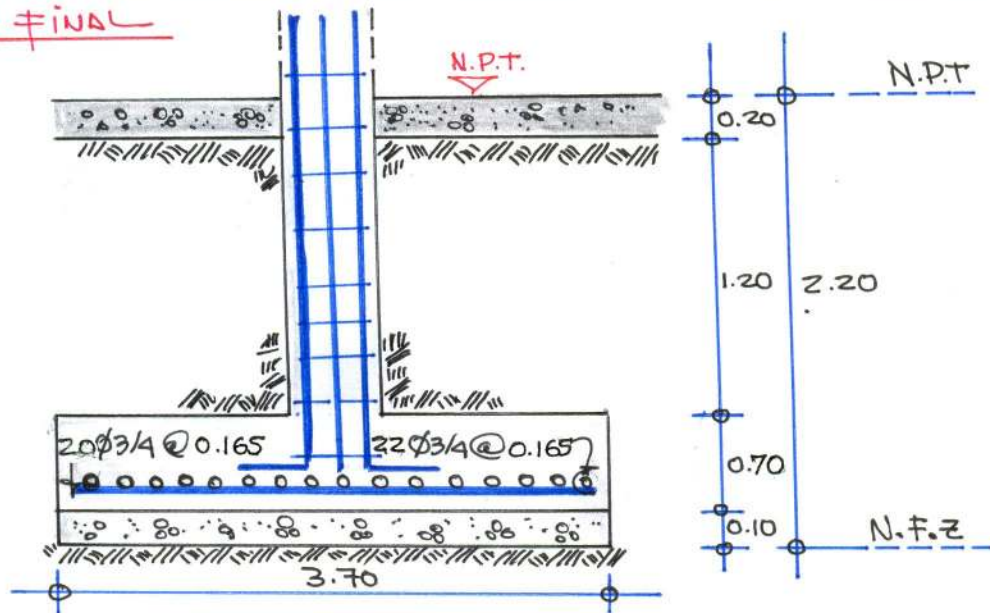
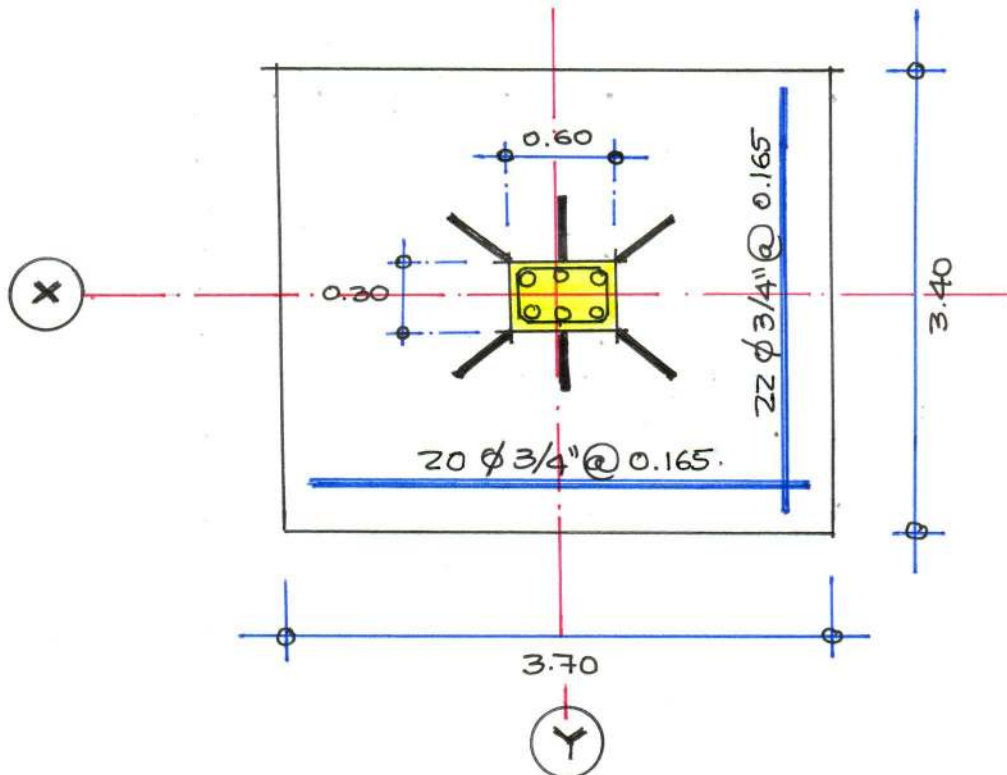
$$P_d = \omega \frac{f_c}{f_y} \Rightarrow P_d = 0.00 \Rightarrow A_{s,d} = P_d b d \Rightarrow A_{s,d} = 7.719 \text{ cm}^2$$

$$A_{s,d} < A_{s,min} \Rightarrow A_{s,d} = 16.90 \text{ cm}^2 \Rightarrow A_{st,d} = A_{s,d} (L) \Rightarrow A_{st,d} = 62.55 \text{ cm}^2$$

$$N = \frac{A_{st,d}}{A_{s\phi}} \Rightarrow N = \frac{62.55}{2.84} \Rightarrow N = 22.027 \text{ UNIDADES} \Rightarrow N = 22 \text{ UNIDADES.}$$

$$S = \frac{L - 2r - \phi - 0.10}{N-1} \Rightarrow \frac{3.70 - 0.15 - 0.01905 - 0.10}{21} \Rightarrow S = 0.16 \text{ m.}$$

∴ Acero transversal: 22 $\phi 3/4 @ 0.16 \text{ m}$

DISEÑO FINALSECCION: X-XPLANTA tipico Z-01DETALLE FINAL DE ZAPATA